Check for updates

The Role of Surgical Lung Biopsy in the Diagnosis of Fibrotic Interstitial Lung Disease

Perspective from the Pulmonary Fibrosis Foundation

Lida P. Hariri[1], Anja C. Roden[2], Jonathan H. Chung[3], Sonye K. Danoff[4], Diana C. Gomez Manjarres[5], Matthew Hartwig[6], Fayez Kheir[7], Christopher King[8], Maryl Kreider[9], David A. Lynch[10], Joshua Mooney[11], Ashok Muniappan[12], Jeffrey L. Myers[13], Luca Paoletti[14], Rishi Raj[15], Zeenat Safdar[16], Sally Suliman[17], Krishna Thavarajah[18], David J. Lederer[19,20], Frederick L. Rudell[20], Pauline Bianchi[20], Barry S. Shea[21], and Brett Ley[22,23]

[1]Department of Pathology and [12]Division of Thoracic Surgery, Department of Surgery, Massachusetts General Hospital and Harvard Medical School, Harvard University, Boston, Massachusetts; [2]Department of Laboratory Medicine and Pathology, Mayo Clinic, Rochester, Minnesota; [3]Department of Radiology, University of Chicago, Chicago, Illinois; [4]Division of Pulmonary and Critical Care Medicine, Johns Hopkins University, Baltimore, Maryland; [5]Division of Pulmonary, Critical Care, and Sleep Medicine, University of Florida, Gainesville, Florida; [6]Division of Cardiovascular and Thoracic Surgery, Department of Surgery, Duke University Medical Center, Duke University, Durham, North Carolina; [7]Division of Pulmonary, Critical Care Medicine and Environmental Medicine, Tulane University, New Orleans, Louisiana; [8]Advanced Lung Disease and Lung Transplant Clinic, Inova Fairfax Hospital, Falls Church, Virginia; [9]Department of Medicine, Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania; [10]Department of Radiology, National Jewish Health, Denver, Colorado; [11]Division of Pulmonary and Critical Care, Department of Medicine, Stanford University, Palo Alto, California; [13]Department of Pathology, Michigan Medicine, Ann Arbor, Michigan; [14]Division of Pulmonary, Critical Care, and Sleep Medicine, College of Medicine, Medical University of South Carolina, Charleston, South Carolina; [15]Division of Pulmonary, Allergy, and Critical Care Medicine, Department of Medicine, Stanford University, Stanford, California; [16]Pulmonary, Critical Care Division, Houston Methodist Hospital and Weill Cornell College of Medicine, Cornell University, Houston, Texas; [17]Department of Medicine, Division of Pulmonary, Critical Care and Sleep Medicine, University of Louisville Health Sciences Center, University of Louisville, Louisville, Kentucky; [18]Department of Internal Medicine, Pulmonary and Critical Care Medicine, Henry Ford Hospital, Detroit, Michigan; [19]Department of Medicine, Columbia University, New York, New York; [20]Pulmonary Fibrosis Foundation, Chicago, Illinois; [21]Division of Pulmonary, Critical Care and Sleep Medicine, Rhode Island Hospital and Warren Alpert Medical School, Brown University, Providence, Rhode Island; [22]Department of Pulmonary and Critical Care Medicine, Kaiser Permanente San Francisco, San Francisco, California; and [23]Department of Medicine, University of California San Francisco, San Francisco, California

ORCID IDs: 0000-0002-4192-5080 (F.K.); 0000-0002-4687-194X (J.M.); 0000-0001-5258-0228 (D.J.L.).

Abstract

Diagnosis of interstitial lung disease (ILD) requires a multidisciplinary discussion approach that includes clinicians, radiologists, and pathologists. Surgical lung biopsy (SLB) is currently the recommended standard in obtaining pathologic specimens for patients with ILD requiring a tissue diagnosis. The increased diagnostic confidence and accuracy provided by microscopic pathology assessment of SLB specimens must be balanced with the associated risks in patients with ILD. This document was developed by the SLB Working Group of the Pulmonary Fibrosis Foundation, composed of a multidisciplinary group of ILD physicians, including pulmonologists, radiologists, pathologists, and thoracic surgeons. In this document, we present an up-to-date literature review of the indications, contraindications, risks, and alternatives to SLB in the diagnosis of fibrotic ILD; outline an integrated approach to the decision-making around SLB in the diagnosis of fibrotic ILD; and provide practical information to maximize the yield and safety of SLB.

Keywords: interstitial lung disease; usual interstitial pneumonitis; idiopathic pulmonary fibrosis; surgical lung biopsy; diagnostic guidelines

(Received in original form September 21, 2020; accepted in final form March 18, 2021)

Correspondence and requests for reprints should be addressed to Lida P. Hariri, M.D., Ph.D., Department of Pathology, Massachusetts General Hospital, 55 Fruit Street, Warren Building 219, Boston, MA 02114. E-mail: lhariri@mgh.harvard.edu.

Ann Am Thorac Soc Vol 18, No 10, pp 1601–1609, Oct 2021
Copyright © 2021 by the American Thoracic Society
DOI: 10.1513/AnnalsATS.202009-1179FR
Internet address: www.atsjournals.org

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026

Diagnosis of interstitial lung disease (ILD) requires a multidisciplinary discussion (MDD) approach that includes input from clinicians, radiologists, and pathologists (1). Pathology plays a key role in the diagnosis of patients with ILD. For patients requiring biopsy for diagnosis, surgical lung biopsy (SLB) is currently the recommended standard in obtaining pathologic specimens for patients with ILD (2). It is generally considered for patients who *1*) lack a confident clinical–radiologic diagnosis after appropriate noninvasive evaluation, *2*) are not at excessively increased risk for postoperative complications, and *3*) are expected to benefit from a change in treatment on the basis of biopsy results (3, 4). Moreover, the pathologic pattern may inform the prognosis of patients with ILD (5, 6). The increased confidence and accuracy of the diagnosis gained from microscopic pathology assessment of SLB specimens must be balanced with the associated risks in patients with ILD.

The aims of this document are to *1*) provide an up-to-date literature review of the indications, contraindications, risks, and alternatives to SLB in the diagnosis of fibrotic ILD; *2*) outline an integrated approach to the decision-making around SLB in the diagnosis of fibrotic ILD; and *3*) provide practical information to maximize the yield and safety of SLB. To achieve these aims, we convened a multidisciplinary group of ILD providers participating in the Pulmonary Fibrosis Foundation Care Center Network, which included 13 pulmonologists, 2 chest radiologists, 3 lung pathologists, and 2 thoracic surgeons with interest and expertise in ILD from 18 centers across the United States.

Indications for SLB

Broadly speaking, SLB is indicated for fibrotic ILD when the results are expected to influence management decisions. The initial treatment approach to newly diagnosed ILDs generally follows one of three basic strategies: *1*) antifibrotic medications currently approved for idiopathic pulmonary fibrosis (IPF) (and recently, nintedanib for scleroderma-associated ILD and progressive fibrosing ILD) (3, 4, 7–9), *2*) immunosuppressive therapy (together with elimination of potentially harmful exposures), or *3*) expectant management. SLB should be considered in cases in which the results are likely to shift diagnostic confidence sufficiently to select one of these three strategies (e.g., IPF vs. fibrotic hypersensitivity pneumonitis [HP], IPF vs. connective tissue disease (CTD)-related ILD [CTD-ILD], IPF vs. other idiopathic interstitial pneumonia). In contrast, SLB may not be appropriate for cases in which the treatment strategy is unlikely to be altered on the basis of SLB results. Examples include cases in which the same immunosuppressive medication would be selected as the treatment choice or when treatment is unlikely to alter the life course (e.g., end-stage disease/appropriate for hospice, need for urgent lung transplantation, or presence of another life-threating comorbidity). A recent randomized clinical trial demonstrated the efficacy of nintedanib for slowing lung function decline in non-IPF progressive fibrotic ILDs (8). However, these data do not obviate the need for an accurate initial diagnosis that can inform treatment strategies, such as those aimed at potentially improving lung function and/or combination therapy with antifibrotic and immunosuppressive approaches.

The decision to pursue SLB depends heavily on the leading or provisional clinical diagnosis and the level of confidence in that diagnosis. Examples of fibrotic ILDs for which SLB may be considered include IPF, fibrotic HP, CTD-ILD, and other exposure-related ILDs.

IPF

For patients with suspected IPF without an identifiable underlying cause of ILD, the high-resolution computed tomography (HRCT) pattern, as outlined by the Fleischner Society (10), provides diagnostic criteria in the context of suspected IPF (Table 1). In addition, the Radiology Working Group of the Pulmonary Fibrosis Foundation developed a practical guide for radiologists on the imaging patterns and reporting of ILD (11). The 2018 American Thoracic Society (ATS)/European Respiratory Society/Japanese Respiratory Society/Asociación Latinoamericana de Tórax guidelines do *not* recommend that patients with a compatible history, typical usual interstitial pneumonia (UIP) pattern at HRCT, and lack of a likely secondary cause for UIP undergo SLB because a confident diagnosis can be made from clinical and radiologic assessment with >90% confidence (1). For patients with suspected IPF with indeterminate and alternative diagnostic patterns on HRCT and a low-confidence clinical diagnosis, SLB is generally recommended but should be considered on a case-by-case basis. For those with probable UIP, biopsy may be considered when the clinical context is uncertain. The guidelines also recommend, when feasible, that the diagnosis of IPF and the decision to perform SLB be considered in the context of MDD because of evidence suggesting improvements in diagnostic confidence provided by MDD when experienced pulmonologists, rheumatologists, thoracic radiologists, and pulmonary pathologists confer. Additional factors to consider in the clinical decision-making are the training and experience of the radiologist, confidence level in the HRCT interpretation, and individual patient characteristics. For example, studies have demonstrated that additional clinical factors, such as male sex and age over 60 years old with a probable UIP pattern on HRCT images read by a thoracic radiologist with high confidence are often considered to have a sufficiently high pretest probability of IPF to preclude the need for SLB for a confident diagnosis of IPF (12–15).

HP

The 2020 ATS/Japanese Respiratory Society/Asociación Latinoamericana de Tórax clinical practice guidelines for the diagnosis of HP categorize HP into nonfibrotic and fibrotic phenotypes on the basis of clinical, radiologic, and pathologic features (16). According to these guidelines, the diagnostic evaluation of possible HP should include *1*) an exposure assessment that includes a thorough clinical history, serum IgG testing against potential eliciting antigens and/or specific inhalational challenge testing in centers with adequate experience; *2*) HRCT evaluation that includes images being acquired in the supine position at deep inspiration and after prolonged expiration as well as review of images by a thoracic radiologist; and *3*) bronchoalveolar lavage (BAL) with lymphocyte analysis (16). A high-confidence diagnosis (80–89% confidence) of HP can be established if an exposure is identified, HRCT shows a typical HP pattern, BAL confirms lymphocytosis, and an MDD was performed. The typical HRCT pattern for fibrotic HP requires a pattern of lung fibrosis characterized by a random distribution both axially and craniocaudally, midlung zone–predominance, or relative sparing in the lower lung zones and at least one abnormality indicative of small airway disease, such as ill-defined, centrilobular nodules and/or ground-glass opacities or mosaic attenuation, a three-density pattern, and/or air-trapping (16).

An HRCT pattern of mosaic attenuation or air-trapping more extensive than the

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026

Table 1. Idiopathic interstitial pneumonia chest HRCT pattern (Fleischner criteria) and 2018 ATS/ERS/JRS/ALAT guideline recommendations for SLB

HRCT Pattern	Typical UIP	Probable UIP	Indeterminate for UIP	Most Consistent with Non-IPF Diagnosis
Distribution	Basal (occasionally diffuse) and subpleural predominant Often heterogeneous	Basal and subpleural predominant Often heterogeneous	Variable or diffuse	Upper- or midlung–predominant fibrosis Peribronchovascular predominance with subpleural sparing
Features	Honeycombing	Reticular pattern (can be superimposed on ground glass) with peripheral traction bronchiectasis or bronchiolectasis	Evidence of fibrosis with some inconspicuous features suggestive of non-UIP pattern	Any of the following: predominant consolidation, extensive pure ground-glass opacity (without acute exacerbation)
	Reticular pattern (can be superimposed on ground glass) with peripheral traction bronchiectasis or bronchiolectasis	Honeycombing is absent		Extensive mosaic attenuation with extensive sharply defined lobular air-trapping on expiration
	Absence of features to suggest an alternative diagnosis	Absence of features to suggest an alternative diagnosis		Diffuse nodules or cysts
Guideline recommendation for SLB	SLB *not* recommended (>90% positive predictive value of UIP pathologic pattern)	Consider SLB (may not be necessary in those with high pretest probability of IPF)	Consider SLB (if no clear etiology is identified after careful clinical and radiologic assessment)	Consider SLB (if no clear etiology is identified after careful clinical and radiologic assessment)

Definition of abbreviations: ALAT = Asociación Latinoamericana de Tórax; ATS = American Thoracic Society; ERS = European Respiratory Society; HRCT = high-resolution computed tomography; IPF = idiopathic pulmonary fibrosis; JRS = Japanese Respiratory Society; SLB = surgical lung biopsy; UIP = usual interstitial pneumonia.
Adapted by permission from References 1 and 10.

reticulation is reported to have a high specificity (>90%) for HP (17). However, it is well known that fibrotic HP can have other HRCT patterns, including typical and probable UIP patterns (10, 17–20). Transbronchial cryobiopsy (TBCB) or SLB are considered in cases in which a confident clinical and radiologic diagnosis cannot be established after an MDD (16), although it should be noted that tissue diagnosis of HP can also be challenging. Although there are histopathologic features that are considered "typical" for HP, such as bronchiolocentric inflammation and/or fibrosis, poorly formed nonnecrotizing granulomas, and chronic interstitial pneumonia, similar features may be seen in the setting of infection or chronic aspiration (21). Conversely, a biopsy showing a predominant histologic pattern of nonspecific interstitial pneumonia or UIP may also be seen in the setting of clinical HP (22, 23). Lung biopsy findings should be considered in the context of clinical, serologic, and HRCT features at MDD.

CTD-ILD

It is generally agreed that SLB is not indicated for patients with ILD and a confirmed diagnosis of systemic CTD if the radiographic pattern is consistent with the underlying disease and its pulmonary manifestations because verification of the histopathologic pattern at SLB has not been demonstrated to alter treatment decisions or the prognosis (24, 25). In addition, the prognosis in patients with CTD-ILD can be predicted without SLB features (26–30). Most patients will be treated with immunosuppressive medications targeting the underlying CTD.

For patients with suspected autoimmune disease who do not meet the criteria for a CTD, the use of SLB may be considered on a case-by-case basis. A framework for evaluating these patients has been proposed on the basis of criteria that include clinical, serologic, and morphologic (HRCT and/or SLB) "domains" under the nomenclature of interstitial pneumonia with autoimmune features (IPAF) (31). However, IPAF criteria are currently considered a research classification tool only. IPAF likely encompasses a heterogeneous group of conditions, including idiopathic (i.e., non–autoimmune-related) and autoimmune etiologies, and has not been proven to predict treatment responses. High confidence of an autoimmune etiology may be provided in some cases on the basis of the combination of clinical, serologic, and HRCT features, and SLB may not be required. In other cases, especially when HRCT features are not strongly suggestive of a pattern more typical of autoimmune disease, SLB may be helpful in determining the likelihood of an autoimmune etiology when considered in the context of clinical, serologic, and HRCT features. However, SLB alone is not sufficient to establish a diagnosis of autoimmune ILD.

Other Exposure-related ILDs

For patients with a suspected drug-induced ILD, pneumoconiosis, or smoking-related ILD, a compatible exposure history and HRCT patterns may be sufficient to make a confident

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026

diagnosis in many cases. The diagnosis centers on the correct identification of the culprit exposure, temporal relationship, and characteristic clinical features and findings at imaging (32). Although, in general, there is no specific diagnostic histologic pattern for drug-induced ILD at SLB, a number of drugs have known patterns of histologic features (33–35). The microscopic pattern at SLB may provide complementary information to the clinical and radiologic information, would increase clinical confidence in the diagnosis, and may be considered on a case-by-case basis.

Risks and Contraindications to SLB

There is no single absolute contraindication to SLB in the diagnosis of ILD. However, there are patient-level risk factors that are associated with increased morbidity and mortality after SLB in patients with ILD. For elective SLB, the risk of short-term (in-hospital) mortality from SLB has been estimated at 1.7% (36, 37), the risk of 30-day mortality has been estimated at 1.5–2.4% (37, 38), and the risk of 90-day mortality has been estimated at 2.8–3.9% (37, 38). For nonelective SLB procedures, short-term mortality may be as high as 16% (36), but some studies report lower 30-day (6.3%) (38) and 90-day (8.8%) (37–39) mortality rates. These risks depend on *1*) the type of procedure (open lung surgery has higher mortality and more complications than video-assisted thoracoscopic surgery [VATS] [38]); *2*) center-level factors (surgical expertise, anesthesia expertise, and SLB volume); and *3*) patient-level factors such as medical comorbidities, older age, the physiologic severity of the ILD, the presence of rapidly progressive disease, the degree of hypoxemia and respiratory failure, and the subsequent disease diagnosis (IPF is associated with higher mortality than other ILDs) (2, 37, 39–43) (Table 2).

The risk of obtaining a nondiagnostic biopsy is about 10%, which can result from either inadequate sampling (*see* below) or nondiagnostic histopathology (1). Other potential complications of SLB include acute exacerbations of the underlying ILD (6.1%, highest risk in IPF), severe bleeding (0.2%), persistent air leak (5.9%), respiratory infection (6.5%), neuropathic or chest-wall pain (4.5%), and delayed wound healing (3.3%) (36). Clinicians and patients should discuss morbidity and mortality related to each patient's specific risk factors as well as technique before the patient undergoes SLB.

Alternatives to SLB

Although SLB remains the diagnostic test of choice for the majority of fibrotic ILDs, other forms of tissue sampling may play a role in the evaluation of patients with ILD. Invasive diagnostic procedures that are alternatives to SLB include BAL fluid analysis, transbronchial forceps biopsies (TBFBs), and TBCBs. In general, these methods provide less information for the diagnosis of fibrotic ILD than does SLB and carry their own specific procedural risks.

BAL

The utility of BAL for the diagnosis of fibrotic ILD remains controversial and is often used as an intermediate test with the aim of increasing or decreasing diagnostic confidence in some cases before deciding whether to proceed to SLB (44, 45). BAL is not indicated as a diagnostic tool in patients with clinical and HRCT features of IPF. In our opinion, BAL may be useful for a newly detected ILD with an unknown cause when HRCT findings indicate probable UIP or are indeterminate for UIP. BAL is recommended in the workup of HP on the basis of recent ATS guidelines and should be considered in all patients with suspected infections, malignancies, or rare ILDs such as pulmonary alveolar proteinosis and eosinophilic lung disease (1, 44, 46).

BAL samples contain both cellular and acellular components of distal bronchioles, which can aid with the diagnosis of a variety of diseases. Fluid sampling should be performed before initiating therapy if possible. The target site should be chosen on the basis of HRCT performed within 6 weeks of BAL to identify regions of abnormality (45). A minimum return of 5 ml is needed for cellular analysis; 10–20 ml is optimal (45). In patients with suspected ILD, a differential cell count should be performed on the fluid and should include macrophages, lymphocytes, neutrophils, eosinophils, and mast cells (44, 45). The remaining BAL sample fluid can be sent for microbiologic/cytologic analysis if clinically indicated. A summary of BAL cellular patterns and their differential diagnosis can be found in the ATS clinical practice guideline on the clinical utility of BAL cellular analysis in ILD (45). The rate of major complications of BAL is $<$0.5%, and the rate of mortality is $<$0.05% (1, 44, 45). The procedure is relatively contraindicated in patients with cardiopulmonary instability or severe bleeding diathesis.

TBFB

The size of TBFB samples is small, typically 2–3 mm in diameter, and sampling of the peripheral lung is difficult. TBFB samples also have significant crush artifacts that makes assessment of lung parenchyma more challenging. The diagnostic yield of TBFB in diffuse parenchymal lung disease is low (approximately 30%) (47–49). Therefore, in our opinion, TBFB should only be considered in lung diseases in which there is a peribronchial and/or central abnormality that is more amenable to TBFB. TBFB is not recommended in ILDs that do not have a peribronchial component because of insufficient tissue size and sampling error (1, 44). TBFB can be diagnostic in select cases, such as when considering a diagnosis of sarcoidosis, HP, eosinophilic pneumonia, organizing pneumonia, pulmonary Langerhans cell histiocytosis, or lymphangioleiomyomatosis (44). The ATS

Table 2. Patient factors associated with increased risk of morbidity and mortality in the setting of surgical lung biopsy in interstitial lung disease

Increased age (e.g., $>$74 yr old)
Immunosuppression
Male sex
Malignancy
Comorbidities
Pulmonary hypertension (PASP $>$ 40 mm Hg)
Decreased pulmonary function (FVC $<$ 50–55%; D_{LCO} $<$ 35–40%)
Hypoxemic respiratory failure (i.e., need for supplemental oxygen)
Nonelective procedure, hospitalized patient, rapidly progressive disease
Corticosteroid therapy (e.g., prednisone $\geqslant$ 20 mg/d)
Frailty, poor functional status
Composite risk categorization based on age, sex, and comorbidities

Definition of abbreviations: D_{LCO} = diffusing lung capacity for carbon monoxide; FVC = forced vital capacity; PASP = pulmonary arterial systemic pressure.
Data are from References 2, 36, 37, 39–43, 76, and 77.

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026

IPF guidelines report that for every 1,000 TBFBs performed, 780 are considered to yield adequate specimens. Of these, diagnoses are made in 360 cases, with the highest yield being found in ILDs with centrilobular disease (1).

When performing TBFB, recent HRCT findings should be used to guide the selection of sampling sites (44). Ideally, at least four to six samples that are 2–3 mm in size should be acquired (44, 50, 51). TBFB should also be performed before the initiation of therapy when possible. The most common complications are pneumothorax (PTX), hypoxemia, bleeding, and arrhythmias. PTX occurs in approximately 10% of cases. Sampling under fluoroscopic guidance can reduce the risk of PTX. Bleeding is the primary limiting factor in obtaining larger samples. The risk of major bleeding is <1% (46). Contraindications include a known bleeding disorder, an international normalized ratio (INR) >1.5, a platelet count of <50,000/μl, and severe hypoxemia (46). Rarely, TBFB can cause an acute exacerbation of IPF (<1%) (46).

TBCB

TBCB uses a bronchoscopic cryoprobe to apply a cooling agent, typically CO_2/nitric oxide under high pressure, which causes lung tissue to adhere to the probe. Subsequently, the cryoprobe with the tissue attached is withdrawn. The role of TBCB in ILD has been heavily debated because of some conflicting literature on its diagnostic accuracy, variable experience levels with this emerging technique, and occasionally ensuing complications (46, 52). When compared with TBFB, TBCB offers the advantage of providing larger tissue samples, a higher percentage of alveolar tissue, and fewer crush artifacts, leading to a greater diagnostic yield (52).

Historically, there has been high heterogeneity in the reported diagnostic yields, which have ranged from 40% to 95.1% based on histopathologic tissue sampling alone (53, 54). The ambiguity surrounding the value of TBCB has led to several landmark trials over the last year, which will likely change the direction of future guidelines surrounding the use of TBCB. A study in 21 patients who underwent both SLB and TBCB in the same procedure demonstrated poor agreement between the diagnosis derived from TBCB findings and the diagnosis derived from SLB findings when the biopsy specimens were reviewed in a blinded manner and without MDD (55). However, more recent larger studies have demonstrated support for the use of TBCB as an alternative to SLB in patients who require lung tissue sampling for an ILD diagnosis (56, 57). The COLDICE (Cryobiopsy vs. Open Lung Biopsy in the Diagnosis of ILD Alliance) trial compared TBCB and SLB samples obtained from the same procedure in 65 patients, and it also incorporated MDD into the blinded assessment of all tissue samples (56). The histopathologic agreement between TBCB and SLB findings was 70.8%, with diagnostic agreement at MDD being 76.9% (κ, 0.62; 0.47–0.78) (56). In 60% of cases, TBCB with MDD provided a high-confidence diagnosis, and in those cases, concordance between the TBCB-derived diagnosis and the SLB-derived diagnosis was high (95%). In the 40% of cases in which TBCB with MDD did not provide a high-confidence diagnosis, concordance with the SLB-derived diagnosis was low. This study suggests that in over half of ILD cases, TBCB combined with MDD may lead to a high-confidence diagnosis without the need for SLB.

Another recent study prospectively enrolled 128 patients with idiopathic interstitial pneumonia to undergo bronchoscopy, BAL, and TBCB, followed by MDD. A confident diagnosis (>90%) was made in 53.9% of cases when TBCB data were incorporated, whereas a confident diagnosis was made in only 22.7% of cases when BAL data alone were used (57). In another retrospective validation study, the distinction between IPF and other ILDs that was made by MDD on the basis of TBCB findings showed a significantly higher 5-year, transplant-free survival rate in patients without a diagnosis of IPF (93%) than in patients with a diagnosis of IPF (68%), which was independent of age, sex, smoking status, comorbidities, pulmonary function, and HRCT patterns (58).

Overall, there appears to be a growing body of literature supporting the use of TBCB for ILD diagnosis, although it remains uncertain where exactly it fits into current diagnostic algorithms. Given the novelty of this technique, there is also a lack of standardization regarding procedure performance, the number and size of specimens to be obtained, and the locations to be used for sampling (1, 52). The overall complication rate is reported to be 23.1%, with the most common complications being bleeding and PTX (46, 53, 59). The mortality rate is reported to be 0.3%, which is lower than is typically reported for SLB (54). Recently, an American College of Chest Physicians (CHEST) guidelines and expert panel report was published on TBCB for the diagnosis of ILDs (60). These guidelines state that although the evidence assessing the safety and utility of TBCB for ILD diagnosis is limited, it may be considered as an alternative to SLB in some circumstances (60).

Emerging Techniques: Genomic Classifiers

A genomic classifier (Envisia) that can be applied to TBFB has recently been introduced for clinical use (61, 62). The classifier has been demonstrated to differentiate UIP from non-UIP histopathologies with high specificity (~86–88%) and appears to increase diagnostic confidence when considered in the context of MDD. However, it cannot establish whether a UIP pattern is due to IPF or an underlying etiology such as CTD or fibrotic HP. The classifier itself also does not provide a clinical diagnosis and lacks other information that is provided by histopathologic analysis of SLB specimens, especially in non-UIP cases, and must therefore be considered in the context of other clinical information, ideally in the setting of MDD. Because of its lower sensitivity for UIP (~63–70%), it is expected that many cases with non-UIP results will still require SLB for a confident diagnosis. Further investigation and clinical experience will be required before the role of the genomic classifier in the diagnosis of ILD can established.

Integrated Approach to Decision-Making to Perform SLB

The overall diagnostic yield of SLB is close to 90% when considered in the context of MDD (1). However, SLB is not necessary to establish a confident diagnosis in most cases in which a confident clinical diagnosis can be established on the basis of noninvasive evaluation, as discussed above. Therefore, before consideration of SLB, a thorough evaluation of the ILD, including a detailed history and physical examination, HRCT, and serologic evaluation for autoimmune disease are critical (1, 10). By using the information obtained from the noninvasive evaluation, the following three questions may help to guide the decision whether to proceed to SLB:

1. What is the leading (or provisional) clinical diagnosis, and what is the level of confidence in this diagnosis?
2. If a confident clinical diagnosis cannot be made, then on the basis of the provisional and alternative diagnoses, how likely are SLB results to alter treatment decisions?

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026

3. If a confident clinical diagnosis cannot be made and SLB results are likely to alter treatment decisions, then is the patient at acceptable risk for complications of SLB?

Role of Multidisciplinary Discussions and Patient Engagement

Consideration of each of the above questions in MDD, or referral to a specialty center for MDD, may aid the clinician in the decision of whether or not to recommend undergoing SLB to an individual patient. Once the clinician has made the decision to recommend SLB, it is important to recognize that decision thresholds are relatively arbitrary and value-based and that individual patients may value benefits and risks differently from clinicians. Therefore, we suggest that patients should be engaged in an informed decision-making discussion about the expected benefits (questions 1 and 2) and risks of SLB (question 3), which should be weighed against the alternative decision of initiating empiric therapy or expectant management on the basis of the provisional or leading diagnosis or diagnoses.

Minimizing the Risks and Maximizing the Yield of SLB

Once the decision has been made to proceed to SLB, it is important to implement strategies to reduce the risk of postoperative complications and maximize the procedure's diagnostic yield. Communication among involved practitioners and institutional protocols implementing these strategies should be considered.

Minimizing the Risks of SLB

SLB can typically be obtained via VATS approaches. Overall, the data indicate that the use of VATS results in lower morbidity and mortality than the use of open lung biopsy, with a lower rate of pleural effusions, PTX, persistent air leak, and hemothorax being shown. VATS also typically has shorter operative times, reduced analgesic use, and shorter pleural drainage durations and hospital stays (63–65). In both open lung biopsy and VATS, anesthesia management can be modified to attempt to reduce the risk of ventilator-associated lung injury (believed to be a cause of acute exacerbations), including minimizing the fraction of inspired oxygen, tidal volume, and peak airway pressure, as well as the amount of perioperative fluid.

One potential method of avoiding ventilator-associated lung injury altogether is to perform nonintubated SLB via VATS techniques, which has been shown to be safe and feasible (66). Although no randomized data comparing intubated with nonintubated SLB are available, some retrospective studies suggest that there are fewer complications with nonintubated approaches (66–68).

Other perioperative considerations include the selective use of stress-dose corticosteroids for those on chronic corticosteroid therapy (preferably after biopsies have been performed) and use of nonnarcotic adjunctive pain strategies postoperatively to improve respiratory mechanics and mobilization, such as nonsteroidal antiinflammatory drugs, acetaminophen, gabapentin, and local nerve blocks.

Maximizing the Yield of SLB

The yield of a SLB depends on optimal sampling to avoid diagnostic bias due to undersampling of the lung tissue and/or disease regions. Considerations for optimal tissue sampling by SLB are summarized in Table 3. Multiple biopsies are needed to assess disease distribution and ensure sufficient sampling for a heterogeneous disease (1, 2, 69). At least one biopsy from two different lobes should be obtained (1, 2, 69). The biopsy location should be correlated with preoperative HRCT scan results to identify areas of high yield for abnormalities and also to avoid areas that only show dense fibrosis and honeycomb changes (1, 2, 70). The tips of the lingula or the right middle lobe of the lung should generally be avoided, as they might show edge artifacts mimicking honeycomb changes (71). Consideration should be given to choosing areas at the interface between fibrotic areas and the preserved lung parenchyma. Sampling error should always be considered, especially if the disease is central. Palpation of the tissue and clamps should be avoided to minimize hemorrhaging and atelectasis.

If the disease is unevenly distributed between the right and left lung at evaluation by HRCT, this should be considered when choosing which side to biopsy. If the disease is symmetrical, no data suggest that one side of the lung is safer to biopsy than the other (69). One study suggested that the right side should be preferred because of the availability of three lobes for tissue sampling (72). Requirements for the size of biopsy specimens are not well studied. On the basis of historical pathologic practice, a biopsy specimen size of $3 \times 2 \times 1$ cm (uninflated) should be sufficient (2). There is no evidence that open SLB or SLB by VATS is superior regarding the yields required for diagnostic accuracy (63, 64, 73–75).

Guidelines for pathologic assessment of SLB are summarized in Table 3. An

Table 3. Considerations for maximizing sampling and pathologic assessment of surgical lung biopsy specimens

Surgical sampling
Multiple biopsy samples from two to three lobes (minimum of one biopsy sample from two different lobes each)
Correlate biopsy location with prior HRCT findings
Avoid sampling of tips of lingula and right middle lobe
Sampling of interface between the area of fibrosis and the architecturally preserved lung parenchyma
Biopsy sample size of approximately $3 \times 2 \times 1$ cm (uninflated)
Do not feel or squeeze specimen and avoid clamps in operating room
Pathologic assessment
Confirm presence of diagnostic material at frozen section laboratory if possible
Fix tissue in formalin for at least 1 h with gentle agitation or formalin injection before sectioning
Cut specimen with sharp knife without squeezing and pushing
H&E: most important stain; possibly add stains for microorganisms, elastic stain, and/or iron on the basis of H&E impression
Possibly seek expert opinion
Multidisciplinary discussion

Definition of abbreviations: H&E = hematoxylin and eosin; HRCT = high-resolution computed tomography.
Data are from References 1, 2, 63, 64, 69, and 71–75.

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026

intraoperative frozen section evaluation of the specimen may be performed to evaluate the adequacy of tissue sampling. In the frozen section laboratory, a single thin piece of tissue is carefully cut from the specimen by using a sharp knife, without pushing or squeezing the specimen. If no abnormal tissue is present or only end-stage fibrotic lung tissue is identified, additional biopsy sampling may be recommended. The final diagnosis should be established on the permanent section slides.

For permanent section evaluation, SLB tissues must be adequately inflated to prevent atelectasis that may hinder pathologic assessment. This can be achieved either by carefully injecting the tissue with formalin (using a small-bore needle and s syringe) or by floating the specimen in formalin with repeated gentle shaking of the container for a minimum of an hour. The entirety of the SLB specimens should be submitted for microscopic assessment. Hematoxylin and eosin staining is the most important stain for microscopic evaluation. Occasionally, a trichrome-stained sample may be evaluated to further assess the fibrotic component. Special stains for microorganisms (Grocott's methenamine silver stain, periodic acid–Schiff stain, acid-fast bacteria stain, or equivalent) should be performed if there is clinical concern for infection or if there are histologic features concerning for or suggestive of potential infection. An elastic stain may be used to assess vascular changes, and an iron stainmay be used if there is concern about asbestos-related lung disease.

We suggest that the interpretation of SLB results be performed by a pathologist with expertise in ILD. The correlation with HRCT and clinical information is also important for pathologists to consider when interpreting SLB results.

Patient Perspective

Mr. Rick Rudell, a patient with IPF and a Pulmonary Fibrosis Foundation Ambassador and Support Group Leader, was invited to review this document. Mr. Rudell underwent an SLB in 2016 and experienced a subsequent acute exacerbation that resulted in a prolonged intensive care unit course. "This is a remarkable document on the subject of surgical lung biopsy, which is very important to the ILD, and particularly the IPF, patient community. If this information had been available in 2015, I may have avoided the SLB. My HRCTs, lung function evaluations for FVC [forced vital capacity] and D_{LCO} [diffusing capacity of the lung for carbon monoxide], and regular clinician visits may have confirmed my diagnosis of IPF with a high degree of certainty. This document will be a great resource for pulmonologists and their patients."

Conclusions

Diagnosis of ILD requires a MDD approach that involves clinicians, radiologists, and pathologists (1). Although pathology plays a key role in the diagnosis of ILD, the increased confidence and accuracy of the diagnosis gained from microscopic pathologic assessment of SLB specimens must be balanced with the associated risks in patients with ILD. ■

Author disclosures are available with the text of this article at www.atsjournals.org.

References

1 Raghu G, Remy-Jardin M, Myers JL, Richeldi L, Ryerson CJ, Lederer DJ, *et al.*; American Thoracic Society; European Respiratory Society; Japanese Respiratory Society; Latin American Thoracic Society. Diagnosis of idiopathic pulmonary fibrosis: an official ATS/ERS/JRS/ALAT clinical practice guideline. *Am J Respir Crit Care Med* 2018;198: e44–e68.

2 Raj R, Raparia K, Lynch DA, Brown KK. Surgical lung biopsy for interstitial lung diseases. *Chest* 2017;151:1131–1140.

3 King TE Jr, Bradford WZ, Castro-Bernardini S, Fagan EA, Glaspole I, Glassberg MK, *et al.*; ASCEND Study Group. A phase 3 trial of pirfenidone in patients with idiopathic pulmonary fibrosis. *N Engl J Med* 2014;370:2083–2092.

4 Richeldi L, du Bois RM, Raghu G, Azuma A, Brown KK, Costabel U, *et al.*; INPULSIS Trial Investigators. Efficacy and safety of nintedanib in idiopathic pulmonary fibrosis. *N Engl J Med* 2014;370:2071–2082.

5 Flaherty KR, Thwaite EL, Kazerooni EA, Gross BH, Toews GB, Colby TV, *et al.* Radiological versus histological diagnosis in UIP and NSIP: survival implications. *Thorax* 2003;58:143–148.

6 Park JH, Kim DS, Park IN, Jang SJ, Kitaichi M, Nicholson AG, *et al.* Prognosis of fibrotic interstitial pneumonia: idiopathic versus collagen vascular disease-related subtypes. *Am J Respir Crit Care Med* 2007;175: 705–711.

7 Distler O, Highland KB, Gahlemann M, Azuma A, Fischer A, Mayes MD, *et al.*; SENSCIS Trial Investigators. Nintedanib for systemic sclerosis-associated interstitial lung disease. *N Engl J Med* 2019;380:2518–2528.

8 Flaherty KR, Wells AU, Cottin V, Devaraj A, Walsh SLF, Inoue Y, *et al.*; INBUILD Trial Investigators. Nintedanib in progressive fibrosing interstitial lung diseases. *N Engl J Med* 2019;381:1718–1727.

9 Wells AU, Flaherty KR, Brown KK, Inoue Y, Devaraj A, Richeldi L, *et al.*; INBUILD Trial Investigators. Nintedanib in patients with progressive fibrosing interstitial lung diseases-subgroup analyses by interstitial lung disease diagnosis in the INBUILD trial: a randomised, double-blind, placebo-controlled, parallel-group trial. *Lancet Respir Med* 2020;8:453–460.

10 Lynch DA, Sverzellati N, Travis WD, Brown KK, Colby TV, Galvin JR, *et al.* Diagnostic criteria for idiopathic pulmonary fibrosis: a Fleischner Society White Paper. *Lancet Respir Med* 2018;6:138–153.

11 Hobbs SB, Chung JH, Leb J, Kaproth-Joslin K, Lynch DA. Practical imaging interpretation in patients suspected of having idiopathic pulmonary fibrosis: official recommendations from the Radiology Working Group of the Pulmonary Fibrosis Foundation. *Radiol Cardiothorac Imaging* 2021; 3:e200279.

12 Brownell R, Moua T, Henry TS, Elicker BM, White D, Vittinghoff E, *et al.* The use of pretest probability increases the value of high-resolution CT in diagnosing usual interstitial pneumonia. *Thorax* 2017;72:424–429.

13 Kim SY, Diggans J, Pankratz D, Huang J, Pagan M, Sindy N, *et al.* Classification of usual interstitial pneumonia in patients with interstitial lung disease: assessment of a machine learning approach using high-dimensional transcriptional data. *Lancet Respir Med* 2015;3:473–482.

14 Salisbury ML, Xia M, Murray S, Bartholmai BJ, Kazerooni EA, Meldrum CA, *et al.* Predictors of idiopathic pulmonary fibrosis in absence of radiologic honeycombing: a cross sectional analysis in ILD patients undergoing lung tissue sampling. *Respir Med* 2016;118:88–95.

15 Chung JH, Chawla A, Peljto AL, Cool CD, Groshong SD, Talbert JL, *et al.* CT scan findings of probable usual interstitial pneumonitis have a high predictive value for histologic usual interstitial pneumonitis. *Chest* 2015; 147:450–459.

16 Raghu G, Remy-Jardin M, Ryerson CJ, Myers JL, Kreuter M, Vasakova M, *et al.* Diagnosis of hypersensitivity pneumonitis in adults: an official ATS/JRS/ALAT clinical practice guideline. *Am J Respir Crit Care Med* 2020; 202:e36–e69.

17 Salisbury ML, Gross BH, Chughtai A, Sayyouh M, Kazerooni EA, Bartholmai BJ, *et al.* Development and validation of a radiological diagnosis model for hypersensitivity pneumonitis. *Eur Respir J* 2018;52: 1800443.

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026

18 Churg A, Sin DD, Everett D, Brown K, Cool C. Pathologic patterns and survival in chronic hypersensitivity pneumonitis. *Am J Surg Pathol* 2009; 33:1765–1770.
19 Vasakova M, Morell F, Walsh S, Leslie K, Raghu G. Hypersensitivity pneumonitis: perspectives in diagnosis and management. *Am J Respir Crit Care Med* 2017;196:680–689.
20 Salisbury ML, Gu T, Murray S, Gross BH, Chughtai A, Sayyouh M, *et al.* Hypersensitivity pneumonitis: radiologic phenotypes are associated with distinct survival time and pulmonary function trajectory. *Chest* 2019;155: 699–711.
21 Cardasis JJ, MacMahon H, Husain AN. The spectrum of lung disease due to chronic occult aspiration. *Ann Am Thorac Soc* 2014;11:865–873.
22 Wang P, Jones KD, Urisman A, Elicker BM, Urbania T, Johannson KA, *et al.* Pathologic findings and prognosis in a large prospective cohort of chronic hypersensitivity pneumonitis. *Chest* 2017;152:502–509.
23 Trahan S, Hanak V, Ryu JH, Myers JL. Role of surgical lung biopsy in separating chronic hypersensitivity pneumonia from usual interstitial pneumonia/idiopathic pulmonary fibrosis: analysis of 31 biopsies from 15 patients. *Chest* 2008;134:126–132.
24 Fischer A, Brown KK, Du Bois RM, Frankel SK, Cosgrove GP, Fernandez-Perez ER, *et al.* Mycophenolate mofetil improves lung function in connective tissue disease-associated interstitial lung disease. *J Rheumatol* 2013;40:640–646.
25 Ryerson CJ, Vittinghoff E, Ley B, Lee JS, Mooney JJ, Jones KD, *et al.* Predicting survival across chronic interstitial lung disease: the ILD-GAP model. *Chest* 2014;145:723–728.
26 Kim EJ, Collard HR, King TE Jr. Rheumatoid arthritis-associated interstitial lung disease: the relevance of histopathologic and radiographic pattern. *Chest* 2009;136:1397–1405.
27 Kim EJ, Elicker BM, Maldonado F, Webb WR, Ryu JH, Van Uden JH, *et al.* Usual interstitial pneumonia in rheumatoid arthritis-associated interstitial lung disease. *Eur Respir J* 2010;35:1322–1328.
28 Morisset J, Vittinghoff E, Lee BY, Tonelli R, Hu X, Elicker BM, *et al.* The performance of the GAP model in patients with rheumatoid arthritis associated interstitial lung disease. *Respir Med* 2017;127:51–56.
29 Cottin V. Significance of connective tissue diseases features in pulmonary fibrosis. *Eur Respir Rev* 2013;22:273–280.
30 Vij R, Strek ME. Diagnosis and treatment of connective tissue disease-associated interstitial lung disease. *Chest* 2013;143:814–824.
31 Fischer A, Antoniou KM, Brown KK, Cadranel J, Corte TJ, du Bois RM, *et al.*; ERS/ATS Task Force on Undifferentiated Forms of CTD-ILD. An official European Respiratory Society/American Thoracic Society research statement: interstitial pneumonia with autoimmune features. *Eur Respir J* 2015;46:976–987.
32 Camus P, Bonniaud P, Fanton A, Camus C, Baudaun N, Foucher P. Drug-induced and iatrogenic infiltrative lung disease. *Clin Chest Med* 2004;25: 479–519, vi.
33 Schwaiblmair M, Behr W, Haeckel T, Märkl B, Foerg W, Berghaus T. Drug induced interstitial lung disease. *Open Respir Med J* 2012;6:63–74.
34 Skeoch S, Weatherley N, Swift AJ, Oldroyd A, Johns C, Hayton C, *et al.* Drug-induced interstitial lung disease: a systematic review. *J Clin Med* 2018;7:356.
35 Roden AC, Camus P. Iatrogenic pulmonary lesions. *Semin Diagn Pathol* 2018;35:260–271.
36 Hutchinson JP, Fogarty AW, McKeever TM, Hubbard RB. In-hospital mortality after surgical lung biopsy for interstitial lung disease in the United States: 2000 to 2011. *Am J Respir Crit Care Med* 2016;193:1161–1167.
37 Hutchinson JP, McKeever TM, Fogarty AW, Navaratnam V, Hubbard RB. Surgical lung biopsy for the diagnosis of interstitial lung disease in England: 1997-2008. *Eur Respir J* 2016;48:1453–1461.
38 Nguyen W, Meyer KC. Surgical lung biopsy for the diagnosis of interstitial lung disease: a review of the literature and recommendations for optimizing safety and efficacy. *Sarcoidosis Vasc Diffuse Lung Dis* 2013; 30:3–16.
39 Fisher JH, Shapera S, To T, Marras TK, Gershon A, Dell S. Procedure volume and mortality after surgical lung biopsy in interstitial lung disease. *Eur Respir J* 2019;53:1801164.
40 Utz JP, Ryu JH, Douglas WW, Hartman TE, Tazelaar HD, Myers JL, *et al.* High short-term mortality following lung biopsy for usual interstitial pneumonia. *Eur Respir J* 2001;17:175–179.
41 Vaszar LT, Larsen BT, Swanson KL, Ryu JH, Tazelaar HD. Diagnostic utility of surgical lung biopsies in elderly patients with indeterminate interstitial lung disease. *Respirology* 2018;23:507–511.
42 Raj R, Brown KK. Mortality related to surgical lung biopsy in patients with interstitial lung disease. The devil is in the denominator. *Am J Respir Crit Care Med* 2016;193:1082–1084.
43 Han Q, Luo Q, Xie JX, Wu LL, Liao LY, Zhang XX, *et al.* Diagnostic yield and postoperative mortality associated with surgical lung biopsy for evaluation of interstitial lung diseases: a systematic review and meta-analysis. *J Thorac Cardiovasc Surg* 2015;149:1394–401, e1.
44 Bradley B, Branley HM, Egan JJ, Greaves MS, Hansell DM, Harrison NK, *et al.*; British Thoracic Society Interstitial Lung Disease Guideline Group, British Thoracic Society Standards of Care Committee; Thoracic Society of Australia; New Zealand Thoracic Society; Irish Thoracic Society. Interstitial lung disease guideline: the British Thoracic Society in collaboration with the Thoracic Society of Australia and New Zealand and the Irish Thoracic Society. *Thorax* 2008;63:v1–v58.
45 Meyer KC, Raghu G, Baughman RP, Brown KK, Costabel U, du Bois RM, *et al.*; American Thoracic Society Committee on BAL in Interstitial Lung Disease. An official American Thoracic Society clinical practice guideline: the clinical utility of bronchoalveolar lavage cellular analysis in interstitial lung disease. *Am J Respir Crit Care Med* 2012;185:1004–1014.
46 Poletti V, Ravaglia C, Gurioli C, Piciucchi S, Dubini A, Cavazza A, *et al.* Invasive diagnostic techniques in idiopathic interstitial pneumonias. *Respirology* 2016;21:44–50.
47 Berbescu EA, Katzenstein ALA, Snow JL, Zisman DA. Transbronchial biopsy in usual interstitial pneumonia. *Chest* 2006;129:1126–1131.
48 Sheth JS, Belperio JA, Fishbein MC, Kazerooni EA, Lagstein A, Murray S, *et al.* Utility of transbronchial vs surgical lung biopsy in the diagnosis of suspected fibrotic interstitial lung disease. *Chest* 2017;151:389–399.
49 Tomassetti S, Cavazza A, Colby TV, Ryu JH, Nanni O, Scarpi E, *et al.* Transbronchial biopsy is useful in predicting UIP pattern. *Respir Res* 2012;13:96.
50 Leslie KO, Gruden JF, Parish JM, Scholand MB. Transbronchial biopsy interpretation in the patient with diffuse parenchymal lung disease. *Arch Pathol Lab Med* 2007;131:407–423.
51 Gilman MJ, Wang KP. Transbronchial lung biopsy in sarcoidosis: an approach to determine the optimal number of biopsies. *Am Rev Respir Dis* 1980;122:721–724.
52 Ussavarungsi K, Kern RM, Roden AC, Ryu JH, Edell ES. Transbronchial cryobiopsy in diffuse parenchymal lung disease: retrospective analysis of 74 cases. *Chest* 2017;151:400–408.
53 Sethi J, Ali MS, Mohananey D, Nanchal R, Maldonado F, Musani A. Are transbronchial cryobiopsies ready for prime time? A systematic review and meta-analysis. *J Bronchology Interv Pulmonol* 2019;26:22–32.
54 Ravaglia C, Bonifazi M, Wells AU, Tomassetti S, Gurioli C, Piciucchi S, *et al.* Safety and diagnostic yield of transbronchial lung cryobiopsy in diffuse parenchymal lung diseases: a comparative study versus video-assisted thoracoscopic lung biopsy and a systematic review of the literature. *Respiration* 2016;91:215–227.
55 Romagnoli M, Colby TV, Berthet JP, Gamez AS, Mallet JP, Serre I, *et al.* Poor concordance between sequential transbronchial lung cryobiopsy and surgical lung biopsy in the diagnosis of diffuse interstitial lung diseases. *Am J Respir Crit Care Med* 2019;199:1249–1256.
56 Troy LK, Grainge C, Corte TJ, Williamson JP, Vallely MP, Cooper WA, *et al.;* Cryobiopsy versus Open Lung Biopsy in the Diagnosis of Interstitial Lung Disease Alliance (COLDICE) Investigators. Diagnostic accuracy of transbronchial lung cryobiopsy for interstitial lung disease diagnosis (COLDICE): a prospective, comparative study. *Lancet Respir Med* 2020; 8:171–181.
57 Hetzel J, Wells AU, Costabel U, Colby TV, Walsh SLF, Verschakelen J, *et al.* Transbronchial cryobiopsy increases diagnostic confidence in interstitial lung disease: a prospective multicentre trial. *Eur Respir J* 2020; 56:1901520.
58 Tomassetti S, Ravaglia C, Wells AU, Cavazza A, Colby TV, Rossi G, *et al.* Prognostic value of transbronchial lung cryobiopsy for the multidisciplinary diagnosis of idiopathic pulmonary fibrosis: a retrospective validation study. *Lancet Respir Med* 2020;8:786–794.
59 Hetzel J, Maldonado F, Ravaglia C, Wells AU, Colby TV, Tomassetti S, *et al.* Transbronchial cryobiopsies for the diagnosis of diffuse parenchymal lung diseases: expert statement from the Cryobiopsy Working Group on

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026

Safety and Utility and a call for standardization of the procedure. *Respiration* 2018;95:188–200.
60 Maldonado F, Danoff SK, Wells AU, Colby TV, Ryu JH, Liberman M, *et al.* Transbronchial cryobiopsy for the diagnosis of interstitial lung diseases: CHEST guideline and expert panel report. *Chest* 2020;157:1030–1042.
61 Pankratz DG, Choi Y, Imtiaz U, Fedorowicz GM, Anderson JD, Colby TV, *et al.* Usual interstitial pneumonia can be detected in transbronchial biopsies using machine learning. *Ann Am Thorac Soc* 2017;14: 1646–1654.
62 Raghu G, Flaherty KR, Lederer DJ, Lynch DA, Colby TV, Myers JL, *et al.* Use of a molecular classifier to identify usual interstitial pneumonia in conventional transbronchial lung biopsy samples: a prospective validation study. *Lancet Respir Med* 2019;7:487–496.
63 Ayed AK, Raghunathan R. Thoracoscopy versus open lung biopsy in the diagnosis of interstitial lung disease: a randomised controlled trial. *J R Coll Surg Edinb* 2000;45:159–163.
64 Bensard DD, McIntyre RC Jr, Waring BJ, Simon JS. Comparison of video thoracoscopic lung biopsy to open lung biopsy in the diagnosis of interstitial lung disease. *Chest* 1993;103:765–770.
65 Fibla JJ, Brunelli A, Cassivi SD, Deschamps C. Aggregate risk score for predicting mortality after surgical biopsy for interstitial lung disease. *Interact Cardiovasc Thorac Surg* 2012;15:276–279.
66 Pompeo E, Rogliani P, Cristino B, Schillaci O, Novelli G, Saltini C. Awake thoracoscopic biopsy of interstitial lung disease. *Ann Thorac Surg* 2013; 95:445–452.
67 Pompeo E, Rogliani P, Atinkaya C, Guerrera F, Ruffini E, Iñiguez-Garcia MA, *et al.*; ESTS awake thoracic surgery working group. Nonintubated surgical biopsy of undetermined interstitial lung disease: a multicentre outcome analysis. *Interact Cardiovasc Thorac Surg* 2019;28:744–750.
68 Kurihara C, Tolly B, DeWolf A, Nader A, Kim S, Odell DD, *et al.* Thoracoscopic lung biopsy under regional anesthesia for interstitial lung disease. *Reg Anesth Pain Med* 2020;45:255–259.
69 Qureshi RA, Ahmed TA, Grayson AD, Soorae AS, Drakeley MJ, Page RD. Does lung biopsy help patients with interstitial lung disease? *Eur J Cardiothorac Surg* 2002;21:621–626. [Discussion, p. 626.]
70 Flaherty KR, Travis WD, Colby TV, Toews GB, Kazerooni EA, Gross BH, *et al.* Histopathologic variability in usual and nonspecific interstitial pneumonias. *Am J Respir Crit Care Med* 2001;164: 1722–1727.
71 Gaensler EA, Carrington CB. Open biopsy for chronic diffuse infiltrative lung disease: clinical, roentgenographic, and physiological correlations in 502 patients. *Ann Thorac Surg* 1980;30:411–426.
72 Blackhall V, Asif M, Renieri A, Civitelli S, Kirk A, Jilaihawi A, *et al.* The role of surgical lung biopsy in the management of interstitial lung disease: experience from a single institution in the UK. *Interact Cardiovasc Thorac Surg* 2013;17:253–257.
73 Miller JD, Urschel JD, Cox G, Olak J, Young JE, Kay JM, *et al.* A randomized, controlled trial comparing thoracoscopy and limited thoracotomy for lung biopsy in interstitial lung disease. *Ann Thorac Surg* 2000;70:1647–1650.
74 Carnochan FM, Walker WS, Cameron EW. Efficacy of video assisted thoracoscopic lung biopsy: an historical comparison with open lung biopsy. *Thorax* 1994;49:361–363.
75 Ferson PF, Landreneau RJ, Dowling RD, Hazelrigg SR, Ritter P, Nunchuck S, *et al.* Comparison of open versus thoracoscopic lung biopsy for diffuse infiltrative pulmonary disease. *J Thorac Cardiovasc Surg* 1993;106: 194–199.
76 Kreider ME, Hansen-Flaschen J, Ahmad NN, Rossman MD, Kaiser LR, Kucharczuk JC, *et al.* Complications of video-assisted thoracoscopic lung biopsy in patients with interstitial lung disease. *Ann Thorac Surg* 2007;83: 1140–1144.
77 Durheim MT, Kim S, Gulack BC, Burfeind WR, Gaissert HA, Kosinski AS, *et al.* Mortality and respiratory failure after thoracoscopic lung biopsy for interstitial lung disease. *Ann Thorac Surg* 2017;104:465–470.

Downloaded from https://academic.oup.com/annalsats/article/18/10/1601/8419215 by Lehrstuhl für Englische Sprache und ihre Didaktik - RWTH Aachen user on 06 March 2026